

✓ 7 SQL and Database

Chapter Introduction

In Chapter 7, we will explore SQL, a programming language to manage large-scale databases. In this notebook, we will use **DuckDB**, a lightweight, serverless database engine renowned for its simplicity and ease of use.

In this chapter, you will learn how to create and manage databases, construct and execute various SQL queries, and manipulate data effectively within the DuckDB environment. Through a series of structured exercises, you will develop the skills necessary to perform essential database operations, such as database creation and data retrieval, updating, deletion, and aggregation.

Goal: Understand the role of database and learn basic skills to use SQL.

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✓ 7.1 Chapter Overview

Keywords: database, RDBMS, table, SQL, MySQL, DuckDB

✓ 7.1.1 What Is a Database

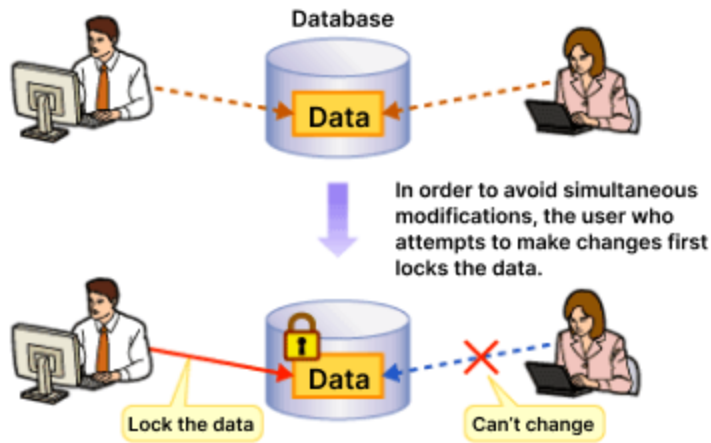
In this section, we will learn about databases.

A **database** is a collection of data that is managed so it can be utilized for various needs later on by using a computer to consolidate large amounts of information in one place.

✓ Why Use Database?

Why do we need databases in the first place? For small datasets that do not require high levels of consistency, Excel might be sufficient. However, data grows by the day, and especially with the rise of IoT, vast amounts of data are now generated in real time. In today's era of **big data**, it is not unusual to see data sources with over 100 million records in a single table, making it extremely difficult to process using Excel alone. By using databases and SQL (the programming language used to manage them), we can efficiently handle large volumes of data. (Today, software like Hadoop and Spark is commonly used for distributed processing of big data.)

Moreover, from a management perspective, Excel has its limitations. By default, anyone can edit an Excel file, and if there is a problem in the data later on, it can be difficult to fix. In Excel, different types of data can be freely entered in columns, and managing them uniformly is challenging. By using a database, data can be managed more easily. Also, to prevent multiple people from editing data simultaneously, databases implement “exclusive control”, which locks the data so others cannot modify it. The figure below illustrates this process.

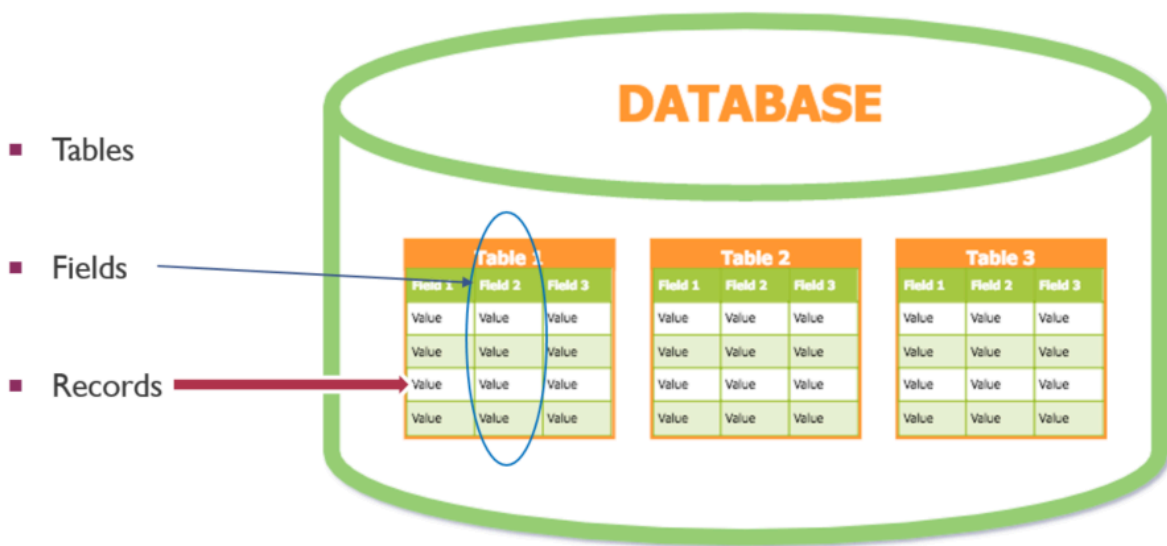


Reference URL : <http://image.itmedia.co.jp/ait/articles/1105/30/r501-01.gif>

✓ Relational Database

In this notebook, we will deal with a type of database called **relational database**. A relational database contains multiple tables, and each table holds different data.

The following figure illustrates the relationship between a database and its tables. Rows in a table are called **records**, and columns are called **columns**, or fields. The tables are connected using **keys**, which are columns that represent a unique identifier of each record and are shared across tables.



Reference URL: <https://hbs-rcs.github.io/sql-novice-survey/fig/RelationalDB.PNG>

A database contains tables, and data is retrieved from these tables. Each table consists of several columns, which can store different types of data, such as numbers, text, or dates. In Excel, you can mix numbers and text in the same column, but in a database table, each column must specify a single data type. Because the data is organized in this manner, it becomes much easier to perform aggregations later on. We will learn how to work with this data in detail as we proceed.

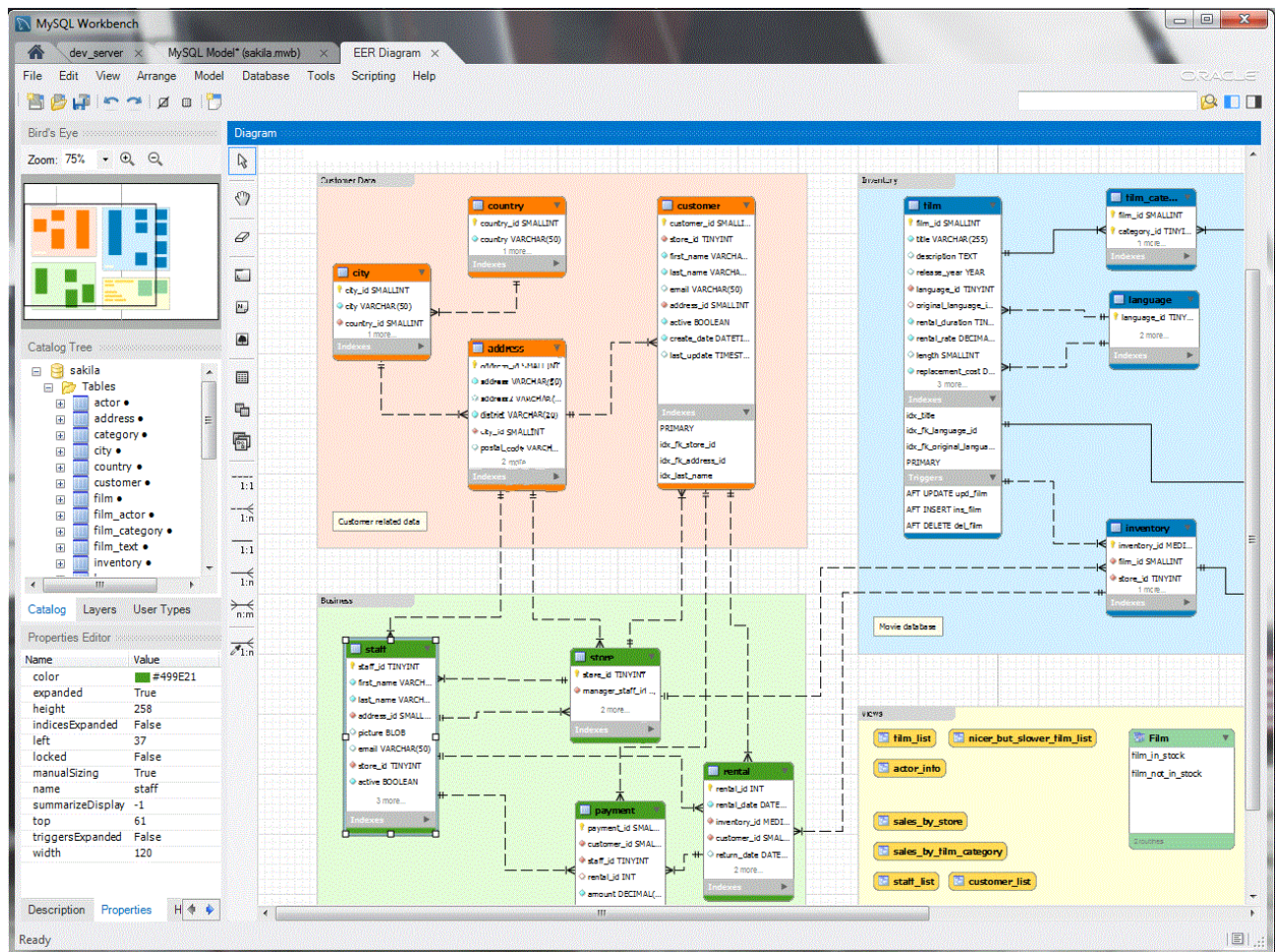
Examples of Relational Database

For instance, a store might manage its daily sales data in a database. Personal customer information, employee salary information, financial transaction data, and website access log data may also be stored in databases. Taking purchase data as an example, you might have entries in a table indicating the date and time a product was sold, the code that identifies the product, the purchaser, the price, and the quantity. These data entries are added on a daily basis, and whenever an aggregation is needed (such as calculating monthly sales), the data is extracted from the database and deleted if no longer necessary.

✓ 7.1.2 RDBMS

The database system we will study in this chapter is called an **RDBMS**, a Relational Database Management System. Major examples include Oracle, Microsoft SQL Server, PostgreSQL, and MySQL.

The figure below shows a conceptual diagram of a RDBMS.



Reference URL:

https://www.mysql.com/common/images/products/MySQL_Workbench_Visual_Design_Windows.gif

In this course, we will use a RDBMS called **DuckDB**.

Note: The following is the documentation of DuckDB, so please consult it as needed.

Reference URL: <https://duckdb.org/>

For those who have never used SQL before, the references below are particularly accessible for beginners. Some involve PHP or other connections, but they are not required to understand this course. If you are interested, feel free to explore them and try creating a simple web service—this is an excellent way to deepen your understanding of databases.

[Reference]

"From the Basics: MySQL, Revised Edition (Kiso kara Series)" by Yumeno Nishizawa (SB Creative)

This concludes our brief introduction to databases. There are many other concepts out there, but let's start with implementation so you can learn how to handle data. You will also learn to use SQL more quickly by actually writing and executing queries.

✓ 7.2 SQL Basics

Keywords: database creation, table, SHOW DATABASES, USE, SHOW TABLES, CREATE, SELECT, WHERE, IN, LIKE, UPDATE, DELETE, ALTER, COUNT, DISTINCT, ORDER, AVG, SUM, LIMIT, GROUP BY, HAVING, UNION, UNION ALL, INNER JOIN, OUTER JOIN, Self-Join, CASE statement, subquery

✓ 7.2.0 Configuration to Use SQL in Google Colab

We will download the required modules.

```
# Download modules for IPython
!pip install jupysql duckdb-engine
```

```
Collecting jupysql
  Downloading jupysql-0.11.1-py3-none-any.whl.metadata (5.9 kB)
Collecting duckdb-engine
  Downloading duckdb_engine-0.17.0-py3-none-any.whl.metadata (8.4 kB)
Requirement already satisfied: prettytable>=3.12.0 in /usr/local/lib/python3.12/
Requirement already satisfied: sqlalchemy in /usr/local/lib/python3.12/dist-pack
Requirement already satisfied: sqlparse in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: ipython-genutils>=0.1.0 in /usr/local/lib/python3
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages
Requirement already satisfied: sqlglot>=11.3.7 in /usr/local/lib/python3.12/dist
Collecting jupysql-plugin>=0.4.2 (from jupysql)
  Downloading jupysql_plugin-0.4.5-py3-none-any.whl.metadata (7.8 kB)
Collecting ploomber-core>=0.2.7 (from jupysql)
  Downloading ploomber_core-0.2.27-py3-none-any.whl.metadata (532 bytes)
Requirement already satisfied: duckdb>=0.5.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: packaging>=21 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages
Collecting posthog>=3.0 (from ploomber-core>=0.2.7->jupysql)
  Downloading posthog-6.6.0-py3-none-any.whl.metadata (6.0 kB)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-package
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: typing-extensions>=4.6.0 in /usr/local/lib/python
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist
Requirement already satisfied: requests<3.0,>=2.7 in /usr/local/lib/python3.12/d
```

```

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packag
Requirement already satisfied: python-dateutil>=2.2 in /usr/local/lib/python3.12
Collecting backoff>=1.10.0 (from posthog>=3.0->ploomber-core>=0.2.7->jupysql)
  Downloading backoff-2.2.1-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: distro>=1.5.0 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-pa
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/d
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/d
Downloading jupysql-0.11.1-py3-none-any.whl (95 kB)
  _____ 95.1/95.1 kB 3.5 MB/s eta 0:00:00
Downloading duckdb_engine-0.17.0-py3-none-any.whl (49 kB)
  _____ 49.7/49.7 kB 3.9 MB/s eta 0:00:00
Downloading jupysql_plugin-0.4.5-py3-none-any.whl (192 kB)
  _____ 192.8/192.8 kB 8.1 MB/s eta 0:00:00
Downloading ploomber_core-0.2.27-py3-none-any.whl (22 kB)
Downloading posthog-6.6.0-py3-none-any.whl (117 kB)
  _____ 117.9/117.9 kB 9.6 MB/s eta 0:00:00
Downloading backoff-2.2.1-py3-none-any.whl (15 kB)
Installing collected packages: backoff, posthog, duckdb-engine, ploomber-core, j
Successfully installed backoff-2.2.1 duckdb-engine-0.17.0 jupysql-0.11.1 jupysql

```

We will load data and set up the database.

```

%load_ext sql
%sql duckdb://

```

Connecting to 'duckdb://'

From this point on, if you place

```
%%sql
```

at the beginning of a cell, you will be able to use SQL.

***Note: Double quotes (") cannot be used. Please use single quotes ('). ***

Additionally, while SQL is not case-sensitive, the following conventions are widely adopted to improve code readability and visibility:

- Write reserved words (SELECT, FROM, WHERE, JOIN, etc.) in uppercase
- Write table names and column names in lowercase
- Use underscores (_) to separate words in multi-word table or column names

✓ 7.2.1 Creating a Database and a Table

**** Note: This includes practice exercises. Please execute them in order as we will use them in subsequent tasks.****

Creating a Database

To create a database, we use `CREATE DATABASE`.

```
CREATE DATABASE database_name;
```

✓ Creating a Table

To create a table, we use `CREATE TABLE`. Contrary to Pandas or Excel, you also need to specify the columns and their data types.

```
CREATE TABLE table_name (  
    column1 INT -- data type: integer  
    column2 VARCHAR(20) -- data type: characters of max length 20  
);
```

You can also specify which columns to use as **primary keys**, etc. By setting as `PRIMARY KEY`, it means that this column is a unique key, therefore disallowing duplicates.

```
CREATE TABLE table_name (  
    column1 INT PRIMARY KEY -- data type: integer, primary key  
    column2 VARCHAR(20) -- data type: characters of max length 20  
);
```

Let's create a table called `emp_list` with an integer-type column named `id` and a text-type column named `name`. We also specify `id` as the primary key.

```
%%sql  
CREATE TABLE emp_list (  
    id INT PRIMARY KEY,  
    name VARCHAR(20)  
);
```

Running query in 'duckdb://'
Count

Retrieving Data

To retrieve a data from a table, we use `SELECT`. When using `SELECT`, you can also combine `FROM` to specify the table and `WHERE` to add any conditions. (See 7.2.2 for conditional retrieval)

To select all columns, we use an asterisk (*).

```
SELECT * FROM table_name;
```

Remember to add a semicolon at the end, which represents the end of a row.

To select certain columns, you would write as follows:

```
SELECT column1, column2 FROM table_name;
```

Let's use `SELECT` on the table we created above. Since the table is empty, you will see an empty table with no records.

```
%%sql
SELECT * FROM emp_list;
```

```
Running query in 'duckdb://'
id name
```

Inserting Data, Rollback

Let's insert data into the table we created. To insert data, we use `INSERT INTO`.

To insert data into specific columns of the table, you would write as follows:

```
INSERT INTO table_name (column1, column2, ...)
VALUES (value1, value2, ...);
```

You can abbreviate the column names if you are inserting into all the columns.

```
INSERT INTO table_name
VALUES (value1, value2, ...);
```

Remember to use single quotation marks when inserting strings.

We insert three records with the following statements.

```
%%sql
INSERT INTO emp_list (id, name) VALUES (1, 'Yamada'); # Please use single quote
```

Running query in 'duckdb://'

Count

```
%%sql
INSERT INTO emp_list (id, name) VALUES (2, 'Tanaka');
```

Running query in 'duckdb://'

Count

```
%%sql
INSERT INTO emp_list (id, name) VALUES (3, 'Sato');
```

Running query in 'duckdb://'

Count

Let's check the table once more.

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'

id name

1 Yamada

2 Tanaka

3 Sato

If you try to insert data for an id that already exists, you will get an error, as shown below.

```
%%sql
INSERT INTO emp_list (id, name) VALUES (1, 'Yamada');
```

Running query in 'duckdb://'

RuntimeError: (duckdb.duckdb.ConstraintException) Constraint Error: Duplicate key
[SQL: INSERT INTO emp_list (id, name) VALUES (1, 'Yamada');]
(Background on this error at: <https://sqlalche.me/e/20/gkpj>)

ROLLBACK clears the interrupted transaction state caused by the error in the previous cell. Without doing this, subsequent code will not run due to errors.

Alternatively, you can disconnect the runtime and re-run everything from the beginning.

```
%%sql
ROLLBACK;
```

Running query in 'duckdb://'
Success

You can check the list of tables with the following SQL statement.

```
%%sql
SELECT name FROM sqlite_master WHERE type='table';
```

Running query in 'duckdb://'
name
emp_list

[Key Points]

CREATE DATABASES [database_name]: Create a database

CREATE TABLE [table_name] [column_name data_type]: Create a table

CREATE TABLE [table_name] [column_name data_type] PRIMARY KEY: Set a primary key

SELECT * FROM table_name: Query all data in the columns of the table

INSERT INTO [table_name] [column_name data_type] [value]: Insert data

✓ <Practice Question 7-1>

Create a new table 'emp_list2.' After creating it, check to make sure the table has been successfully created. You may choose any column names and data types.

```
%%sql
CREATE TABLE emp_list2 (
    id INT PRIMARY KEY,
    dinosaurs VARCHAR(20),
    period VARCHAR(20)
);
```

Running query in 'duckdb://'
Count

```
%%sql  
SELECT * FROM emp_list2;
```

Running query in 'duckdb://'
id dinosaurs period

Start coding or [generate](#) with AI.

Double-click (or enter) to edit

✓ <Practice Question 7-2>

Add data to the new table 'emp_list2' you created in Practice Question 7-1. After adding the data, verify that the table indeed contains the new records.

Start coding or [generate](#) with AI.

✓ 7.2.2 Searching, Updating, and Deleting Data and Columns

✓ Searching Data

Next, let's try extracting data under certain conditions. We use **SELECT** statement with **WHERE** clause for conditional retrieval.

```
SELECT column1, ... FROM table_name WHERE condition;
```

In the example below, we are extracting the record for the person whose **id** is 2.

```
%%sql  
SELECT * FROM emp_list WHERE id=2;
```

Running query in 'duckdb://'
id name
2 Tanaka

There are several useful operators apart from basic conditional operators, such as equal to (`=`), greater than (`>`), or less than (`<`).

By using `IN`, we can extract records that match several possible values.

The example below extracts records where the `id` is either 1 or 2.

```
%%sql
SELECT * FROM emp_list WHERE id IN (1,2);
```

Running query in 'duckdb://'

```
id name
1 Yamada
2 Tanaka
```

By using `LIKE`, we can perform partial (fuzzy) matching. One wildcard is `%`, which represents any number of characters, including zero.

```
SELECT column1, ... FROM table_name
WHERE columnN LIKE 'A%'; -- starts with A
```

```
SELECT column1, ... FROM table_name
WHERE columnN LIKE '%A%'; -- contains A
```

```
SELECT column1, ... FROM table_name
WHERE columnN LIKE '%A'; -- ends with A
```

In this example, we search for any `name` that starts with S.

```
%%sql
SELECT * FROM emp_list WHERE name LIKE 'S%';
```

Running query in 'duckdb://'

```
id name
3 Sato
```

There is also `BETWEEN`, which extracts data between the specified range. See documentation for more details.

[Try It]

Experiment with changing the conditions above to search the data in various ways. For instance, how would you search for people whose id is not 1, or for people whose name ends with 'a'?

✓ Updating Data

Up to now, we have been extracting existing data, but let's try modifying the data. To modify a record, we use `UPDATE`. The syntax is as follows:

```
UPDATE table_name
SET column1 = value1, column2 = value2, ...
WHERE condition;
```

Note that if you forget to add the conditions, all the records will be overwritten.

We will use `UPDATE` to change the record for the person whose `id` is 3.

```
%%sql
UPDATE emp_list
SET name = 'Suzuki'
WHERE id = 3;
```

Running query in 'duckdb://'
Count

Check whether the update took place.

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'
id name
1 Yamada
2 Tanaka
3 Suzuki

✓ Deleting Data

Next, let's discuss how to delete data. We'll prepare sample data (`id=4`) to apply deletion.

```
%%sql
INSERT INTO emp_list (id, name) VALUES (4, 'Dummy');
```

Running query in 'duckdb://'
Count

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'
id name
1 Yamada
2 Tanaka
3 Suzuki
4 Dummy

Use **DELETE** to remove data. The syntax is as follows:

```
DELETE FROM table_name
WHERE condition;
```

Here, we will remove the person whose **id** is 4 that we prepared just now.

```
%%sql
DELETE FROM emp_list
WHERE id = 4;
```

Running query in 'duckdb://'
Count

For caution's sake, verify that the record is indeed gone.

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'
id name
1 Yamada
2 Tanaka
3 Suzuki

✓ Modify Columns

Next, let's add new columns.

To modify columns of existing table, we use `ALTER TABLE`. To add columns, we write as follows:

```
ALTER TABLE table_name ADD column_name datatype;
```

You can also use `ALTER TABLE` to delete or rename columns as well. See documentation for more details.

We will add the person's class and age.

```
%%sql
ALTER TABLE emp_list ADD class INT;
```

Running query in 'duckdb://'
Success

```
%%sql
ALTER TABLE emp_list ADD age INT;
```

Running query in 'duckdb://'
Success

Check the table with the newly added columns. The `class` and `age` columns are still empty (NULL) because no data has been entered yet.

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'
id name class age
1 Yamada None None
2 Tanaka None None
3 Suzuki None None

Use UPDATE to update the data.

```
%%sql
UPDATE emp_list
SET class=1,age=13
WHERE id=3;
```

Running query in 'duckdb://'
Count

The record for the person whose **id** is 3 has now been updated with **class** and **age**.

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'
id name class age
 1 Yamada None None
 2 Tanaka None None
 3 Suzuki 1 13

Let's continue and update the data for the other people as well.

```
%%sql
UPDATE emp_list
SET class=1,age=14
WHERE id=1;
```

Running query in 'duckdb://'
Count

```
%%sql
UPDATE emp_list
SET class=2,age=13
WHERE id=2;
```

Running query in 'duckdb://'
Count

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'
id name class age
 1 Yamada 1 14
 2 Tanaka 2 13
 3 Suzuki 1 13

Now let's add data for some new people. Previously, we ran INSERT statements one at a time. Here, we will update multiple records at once with the following command.

```
%%sql
INSERT INTO emp_list (id,name,class,age)
VALUES (4,'Kato',2,15),(5,'Ito',3,12),(6,'Takeuchi',2,16),(7,'Kimura',3,11),(8,
```

Running query in 'duckdb://'
Count

Check the data.

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'

id	name	class	age
1	Yamada	1	14
2	Tanaka	2	13
3	Suzuki	1	13
4	Kato	2	15
5	Ito	3	12
6	Takeuchi	2	16
7	Kimura	3	11
8	Sato	1	14

[Key Points]

SELECT * FROM [table_name] WHERE condition: Narrow down data by condition

IN: Condition match

LIKE: Fuzzy search

DELETE FROM [table_name]: Delete data (note: use WHERE clause to narrow down if needed)

ALTER [table_name] ADD [column_name data_type]: Add a column to the table

UPDATE [table_name] SET : Update data

✓ <Practice Question 7-3>

In the 'emp_list' table mentioned above, extract the record for the person whose id is 4.

```
%%sql
SELECT * FROM emp_list WHERE id=4;
```

Running query in 'duckdb://'

id	name	class	age
4	Kato	2	15

✓ <Practice Question 7-4>

In the same 'emp_list' table, update the class for the person whose id is 8 to 7. After verifying with a SELECT statement that the update was successful, revert the class for the person whose id is 8 back to 1.

```
%%sql
UPDATE emp_list
SET class=7
WHERE id=8;
```

Running query in 'duckdb://'

Count

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'

id	name	class	age
1	Yamada	1	14
2	Tanaka	2	13
3	Suzuki	1	13
4	Kato	2	15
5	Ito	3	12
6	Takeuchi	2	16
7	Kimura	3	11
8	Sato	7	14

```
%%sql
UPDATE emp_list
SET class=1
WHERE id=8;
```

Running query in 'duckdb://'

Count

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'

id	name	class	age
1	Yamada	1	14
2	Tanaka	2	13
3	Suzuki	1	13
4	Kato	2	15
5	Ito	3	12
6	Takeuchi	2	16
7	Kimura	3	11
8	Sato	1	14

Start coding or [generate](#) with AI.

✓ <Practice Question 7-5> (※Mandatory: will be used in subsequent problems)

In the same 'emp_list' table, add a new column called 'height.' Then update it so that the people whose ids range from 1 to 4 have a height of 150, those whose ids are 5 to 6 have a height of 155, and those whose ids are 7 to 8 have a height of 160.

```
%%sql
ALTER TABLE emp_list ADD COLUMN height INT;
```

Running query in 'duckdb://'
Success

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'

id	name	class	age	height
1	Yamada	1	14	None
2	Tanaka	2	13	None
3	Suzuki	1	13	None
4	Kato	2	15	None
5	Ito	3	12	None
6	Takeuchi	2	16	None
7	Kimura	3	11	None
8	Sato	1	14	None

```
%%sql
UPDATE emp_list
SET height=150
WHERE id IN (1,2,3,4);
```

Running query in 'duckdb://'
Count

```
%%sql
UPDATE emp_list
SET height=150
WHERE id<=4;
```

Running query in 'duckdb://'
Count

```
%%sql
UPDATE emp_list
SET height=150
WHERE id BETWEEN 1 AND 4;
```

Running query in 'duckdb://'
Count

Double-click (or enter) to edit

```
%%sql
UPDATE emp_list
SET height=155
WHERE id IN (5,6);
```

Running query in 'duckdb://'
Count

```
%%sql
UPDATE emp_list
SET height=160
WHERE id IN (7,8);
```

Running query in 'duckdb://'
Count

```
%%sql
SELECT * FROM emp_list;
```

Running query in 'duckdb://'

id	name	class	age	height
1	Yamada	1	14	150
2	Tanaka	2	13	150
3	Suzuki	1	13	150
4	Kato	2	15	150
5	Ito	3	12	155
6	Takeuchi	2	16	155
7	Kimura	3	11	160
8	Sato	1	14	160

```
%%sql SELECT * FROM emp_list;
```

Double-click (or enter) to edit

```
%%sql
```

```
INSERT INTO emp_list (id, name)
```

```
VALUES (9, 'suzuki');
```

Running query in 'duckdb://'

Count

```
%%sql
```

```
SELECT * FROM emp_list;
```

Running query in 'duckdb://'

id	name	class	age	height
1	Yamada	1	14	150
2	Tanaka	2	13	150
3	Suzuki	1	13	150
4	Kato	2	15	150
5	Ito	3	12	155
6	Takeuchi	2	16	155
7	Kimura	3	11	160
8	Sato	1	14	160
9	suzuki	None	None	None

```
%%sql
```

```
SELECT * FROM emp_list;
```

Running query in 'duckdb://'

id	name	class	age	height
1	Yamada	1	14	150
2	Tanaka	2	13	150
3	Suzuki	1	13	150
4	Kato	2	15	150
5	Ito	3	12	155
6	Takeuchi	2	16	155
7	Kimura	3	11	160
8	Sato	1	14	160
9	suzuki	None	None	None

```
%%sql
```

```
SELECT * FROM emp_list WHERE name LIKE 's%';
```

Running query in 'duckdb://'

id	name	class	age	height
9	suzuki	None	None	None

Double-click (or enter) to edit

Double-click (or enter) to edit

✓ 7.2.3 Data Aggregation, Arithmetic, and Sorting

Next, we will look at aggregation using SQL.

✓ Counting Data

One of the first things we often want to know when looking at data is the number of records.

In SQL, we use `COUNT()` function to count the number of records.

```
SELECT COUNT(*) FROM table_name;
```

Below, we use `COUNT` to determine how many rows there are in `emp_list`.

```
%%sql
```

```
SELECT COUNT(*) FROM emp_list;
```

```
Running query in 'duckdb://'  
count_star()  
9
```

You can assign names using `AS`. This is called an alias. Let's name the column `cnt`.

```
%%sql  
SELECT COUNT(*) AS cnt FROM emp_list;
```

```
Running query in 'duckdb://'  
cnt  
9
```

Next, let's use `COUNT` to count the different types of data. Specifically, we previously saw that there are 8 records, but how many different classes are there? By combining `COUNT` and `DISTINCT`, we can count the number of unique items.

```
%%sql  
SELECT COUNT(DISTINCT class) AS classCnt FROM emp_list;
```

```
Running query in 'duckdb://'  
classCnt  
3
```

✓ Calculating Using Data

Next, let's calculate the average. We use `AVG()` function to calculate it.

The following example calculates the average age.

```
%%sql  
SELECT AVG(age) AS avgAge FROM emp_list;
```

```
Running query in 'duckdb://'  
avgAge  
13.5
```

Using the `SUM()` function would give us the total value of a column.

Here, let's confirm that the average age we calculated above matches with taking the sum and dividing by the total number of records.

```
%%sql
SELECT CAST(SUM(age) AS REAL) / CAST(COUNT(*) AS REAL) AS avgAge FROM emp_list;
```

Running query in 'duckdb://'

```
avgAge
12.0
```

You will find that it matches the previous result. As you can see, arithmetic operations can be done in the SELECT statement itself.

We can also find the maximum and minimum ages as follows. We see that the youngest is 11 years old and the oldest is 16.

```
%%sql
SELECT MIN(age) AS minAge, MAX(age) AS maxAge FROM emp_list;
```

Running query in 'duckdb://'

```
minAge maxAge
11      16
```

✓ Sorting Data

To sort a table, we use `ORDER BY`.

```
SELECT column1, ... FROM table_name
ORDER BY columnN; -- default is in ascending order
```

To sort in descending order, we add `DESC` argument at the end.

```
SELECT column1, ... FROM table_name
ORDER BY columnN DESC; -- in descending order
```

Note that for strings, `ORDER BY` will sort in alphabetical order.

Let's sort in descending order based on the `id`.

```
%%sql
SELECT * FROM emp_list ORDER BY id DESC;
```

Running query in 'duckdb://'

	id	name	class	age	height
9	suzuki	None	None	None	
8	Sato	1	14	160	
7	Kimura	3	11	160	
6	Takeuchi	2	16	155	
5	Ito	3	12	155	
4	Kato	2	15	150	
3	Suzuki	1	13	150	
2	Tanaka	2	13	150	
1	Yamada	1	14	150	

Now, let's sort by **age** in ascending order (from smallest to largest). The default is ascending.

```
%%sql
```

```
SELECT * FROM emp_list ORDER BY age;
```

Running query in 'duckdb://'

	id	name	class	age	height
7	Kimura	3	11	160	
5	Ito	3	12	155	
2	Tanaka	2	13	150	
3	Suzuki	1	13	150	
1	Yamada	1	14	150	
8	Sato	1	14	160	
4	Kato	2	15	150	
6	Takeuchi	2	16	155	
9	suzuki	None	None	None	

Then, let's try sorting by two columns: **class** and **age**. The column listed first has priority (in this case, **class**).

```
%%sql
```

```
SELECT * FROM emp_list ORDER BY class, age;
```

Running query in 'duckdb://'

id	name	class	age	height
3	Suzuki	1	13	150
1	Yamada	1	14	150
8	Sato	1	14	160
2	Tanaka	2	13	150
4	Kato	2	15	150
6	Takeuchi	2	16	155
7	Kimura	3	11	160
5	Ito	3	12	155
9	suzuki	None	None	None

✓ Side Note 1: Limiting Number of Rows to Return

Using `LIMIT` allows you to limit the number of rows returned. When working with large volumes of data, you can limit the display count by using `LIMIT`.

```
%%sql
```

```
SELECT * FROM emp_list LIMIT 3;
```

Running query in 'duckdb://'

id	name	class	age	height
1	Yamada	1	14	150
2	Tanaka	2	13	150
3	Suzuki	1	13	150

✓ Side Note 2: Searching Data with Multiple Conditions

Next, let's narrow down records with multiple conditions. Use `AND` to specify simultaneous conditions.

```
%%sql
```

```
SELECT * FROM emp_list WHERE class=2 AND age>=15;
```

Running query in 'duckdb://'

id	name	class	age	height
4	Kato	2	15	150
6	Takeuchi	2	16	155

Next, we use `OR`. This is used when a record meets at least one of the specified conditions.

```
%%sql
SELECT * FROM emp_list WHERE class=1 OR age>=15;
```

Running query in 'duckdb://'

id	name	class	age	height
1	Yamada	1	14	150
3	Suzuki	1	13	150
4	Kato	2	15	150
6	Takeuchi	2	16	155
8	Sato	1	14	160

[Key Points]

COUNT: Count the number of records

AS: Alias (naming)

DISTINCT: Eliminate duplicates

AVG: Average value

MAX: Maximum value

MIN: Minimum value

ORDER BY column_name: Sort in ascending order (use DESC after the column name for descending)

✓ <Practice Question 7-6>

In the same 'emp_list' table mentioned above, how many different ages are there? Name the number of the count as 'ageCnt'.

```
%%sql
SELECT COUNT(DISTINCT age) AS ageCnt FROM emp_list;
```

Running query in 'duckdb://'

ageCnt
6

✓ <Practice Question 7-7>

Find the height of the tallest and shortest persons. Show these values as 'minHeight' and 'maxHeight'.

Start coding or [generate](#) with AI.

✓ <Practice Question 7-8>

Extract the records for those whose height is 155 or more and whose class is 3.

Start coding or [generate](#) with AI.

✓ 7.2.4 Aggregating by Group

Now let's use SQL to aggregate data by group. This is similar to the pivot table feature in Excel. In SQL, we use `GROUP BY` statement. The syntax is as follows:

```
SELECT column1, ...  
FROM table_name  
GROUP BY column_name(s)
```

`GROUP BY` is often used with aggregation functions such as `COUNT()`, `AVG()`, `MAX()`, `MIN()`, etc.

Here, we will use the `class` as the grouping axis. The following command counts how many people are in each `class`.

```
%%sql  
SELECT class, COUNT(*) AS num FROM emp_list GROUP BY class;
```

Running query in 'duckdb://'

class	num
-------	-----

1	3
---	---

2	3
---	---

3	2
---	---

It is also possible to calculate averages for each group. Similarly, you can find minimums and maximums.

```
%%sql  
SELECT class, COUNT(*) AS num, AVG(age) AS avgAge FROM emp_list GROUP BY class;
```

Running query in 'duckdb://'

	class	num	avgAge
1	3		13.666666666666666
2	3		14.666666666666666
3	2		11.5

We can also filter by conditions.

To filter **after** grouping, you use **HAVING**.

```
%%sql
```

```
SELECT class, AVG(age) AS avgAge FROM emp_list GROUP BY class HAVING avgAge>13;
```

Running query in 'duckdb://'

	class	avgAge
1		13.666666666666666
2		14.666666666666666

To filter **before** grouping, you use **WHERE**.

```
%%sql
```

```
SELECT class, AVG(age) AS avgAge FROM emp_list WHERE age > 13 GROUP BY class;
```

Running query in 'duckdb://'

	class	avgAge
1		14.0
2		15.5

See the difference in where **HAVING** or **WHERE** is inserted.

Now let's use two axes for aggregation. We will count records by **class** × **age**.

```
%%sql
```

```
SELECT class, age, COUNT(*) AS num FROM emp_list GROUP BY class, age;
```

Running query in 'duckdb://'

class age num

1	13	1
1	14	2
2	13	1
2	15	1
2	16	1
3	11	1
3	12	1

You can also specify which columns to **GROUP BY** by specifying their column numbers. Note that this may be confusing, especially when dealing with larger databases. For anything beyond a simple summary, specifying column names is recommended.

```
%%sql
```

```
SELECT class, age, COUNT(*) AS num FROM emp_list GROUP BY 1, 2;
```

Running query in 'duckdb://'

class age num

1	13	1
1	14	2
2	13	1
2	15	1
2	16	1
3	11	1
3	12	1

[Key Points]

GROUP BY: Set an axis

HAVING: Set a condition (after grouping)

So far, we have seen many clauses for querying (e.g., WHERE, GROUP BY, HAVING).

To help you remember the clauses and the correct order to apply them, there is a classical mnemonic.

[Key Points] The order of the querying clauses:

SELECT ... FROM ... WHERE ... GROUP BY ... HAVING ... ORDER BY ...

Mnemonic: **So Few Workers Go Home On** time.

✓ <Practice Question 7-9>

In the 'emp_list' table, find the average height for each class. Name the value as 'avgHeight'.

Start coding or [generate](#) with AI.

✓ <Practice Question 7-10>

In addition, for each class, find the number of people, the shortest person's height (as 'minHeight'), and the tallest person's height (as 'maxHeight').

Start coding or [generate](#) with AI.

✓ <Practice Question 7-11>

Now filter for people older than 13 and calculate the same items listed above.

Start coding or [generate](#) with AI.

✓ 7.2.5 Using Multiple Tables

Next, let's operate on data by using multiple tables. To prepare, we will create a table. Although we could create one using the same method we learned previously, it can be a bit tedious, so we will "copy" an existing table. Below, we use CREATE to name the new table and copy the entire table structure.

We will not use the table created in the previous practice questions, so we will delete it.

```
%%sql
DROP TABLE IF EXISTS emp_list2;
```

Running query in 'duckdb://'
Success

To copy an existing table, we combine **COMBINE TABLE** with **AS**.

```
CREATE TABLE new_table_name AS
SELECT column1, ...
```

```
FROM table_name  
WHERE condition;
```

The records will also be copied from the existing table. To delete all records, we can use a condition that does not match with any record in the existing table.

For example, `class=100000` does not exist, so there will be zero records.

```
%%sql  
CREATE TABLE emp_list2 AS SELECT * FROM emp_list WHERE class=10000;
```

Running query in 'duckdb://'

Count

Check it.

```
# Check the data  
%%sql  
SELECT * FROM emp_list2;
```

Running query in 'duckdb://'

id name class age height

Next, we will add data to the new table. Note that the first record is intentionally the same as an existing record in the `emp_list` table (Yamada).

```
%%sql  
INSERT INTO emp_list2 (id, name, class, age, height)  
VALUES (1, 'Yamada', 1, 14, 150),  
      (9, 'Nagata', 4, 11, 160),  
      (10, 'Sugino', 2, 13, 150),  
      (11, 'Takayama', 1, 14, 155),  
      (12, 'John', 3, 16, 165);
```

Running query in 'duckdb://'

Count

```
%%sql  
SELECT * FROM emp_list2;
```

Running query in 'duckdb://'

id	name	class	age	height
1	Yamada	1	14	150
9	Nagata	4	11	160
10	Sugino	2	13	150
11	Takayama	1	14	155
12	John	3	16	165

Now, let's combine the data.

✓ UNION Operator

To concatenate multiple results of `SELECT` statements, we can use `UNION`.

```
SELECT column_name(s) FROM table1
UNION -- excludes duplicates
SELECT column_name(s) FROM table2
```

```
SELECT column_name(s) FROM table1
UNION ALL -- allows duplicates
SELECT column_name(s) FROM table2
```

As noted above, `UNION` excludes duplicate records, whereas `UNION ALL` allows duplicates.

```
%%sql
SELECT * FROM emp_list
UNION
SELECT * FROM emp_list2;
```

Running query in 'duckdb://'

id	name	class	age	height
5	Ito	3	12	155
6	Takeuchi	2	16	155
1	Yamada	1	14	150
9	Nagata	4	11	160
10	Sugino	2	13	150
11	Takayama	1	14	155
12	John	3	16	165
2	Tanaka	2	13	150
3	Suzuki	1	13	150
4	Kato	2	15	150

Truncated to [displaylimit](#) of 10.

If we use `UNION ALL`, you will see two records for `id=1`.

```
%%sql
SELECT * FROM emp_list
UNION ALL
SELECT * FROM emp_list2;
```

Running query in 'duckdb://'

id name class age height

1	Yamada	1	14	150
2	Tanaka	2	13	150
3	Suzuki	1	13	150
4	Kato	2	15	150
5	Ito	3	12	155
6	Takeuchi	2	16	155
7	Kimura	3	11	160
8	Sato	1	14	160
1	Yamada	1	14	150
9	Nagata	4	11	160

Truncated to [displaylimit](#) of 10.

✓ JOIN Operator

Next, let's create a test score table and join it with the 'emp_list' table.

```
%%sql
CREATE TABLE score
(id INT PRIMARY KEY, math INT, english INT, science INT);
```

Running query in 'duckdb://'

Count

Insert sample data into the 'score' table.

```
%%sql
INSERT INTO score (id, math, english, science)
VALUES (1,10,10,40),(2,NULL,60,100),(3,30,40,60),
(5,25,31,10),(6,50,30,10),(8,90,100,90),
(9,67,71,50),(11,100,40,88),(15,76,21,13);
```

Running query in 'duckdb://'

Count

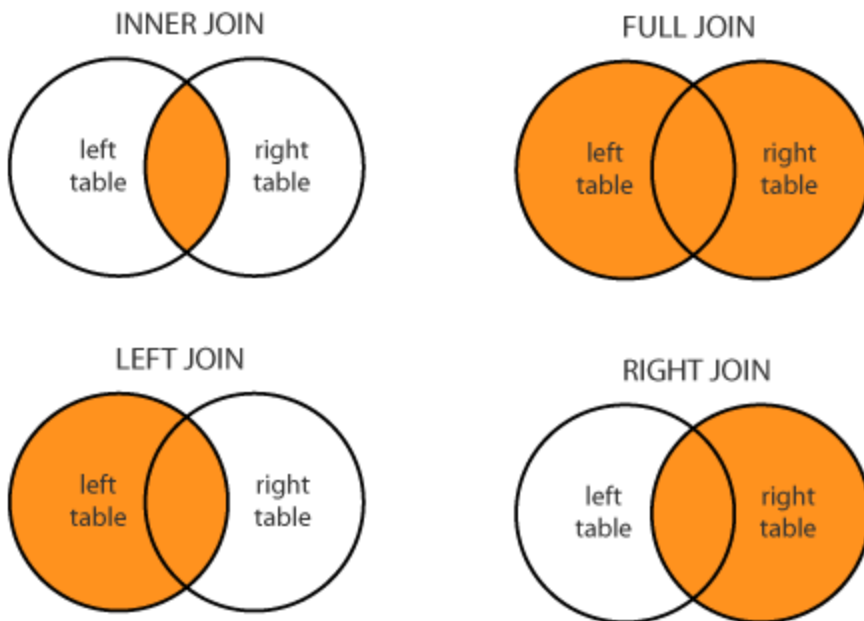
Check to make sure the data is there.

```
%%sql
SELECT * FROM score;
```

Running query in 'duckdb://'

	id	math	english	science
1	10	10	40	
2	None	60	100	
3	30	40	60	
5	25	31	10	
6	50	30	10	
8	90	100	90	
9	67	71	50	
11	100	40	88	
15	76	21	13	

In SQL, we use the **JOIN** operator to join multiple tables. There are four types of JOINS, **(INNER) JOIN**, **LEFT (OUTER) JOIN**, **RIGHT (OUTER) JOIN**, and **FULL (OUTER) JOIN**.



Reference: <https://www.quora.com/Why-are-SQL-joins-so-tough>

The syntax of joining is as follows:

```
SELECT column_name(s)
FROM table1
JOIN table2 -- join type you want to use
ON table1.column_name = table2.column_name;
```

You can also assign simple aliases to the table names.

```
SELECT column_name(s)
FROM table1 t1 -- assign table1 as t1
JOIN table2 t2 -- assign table2 as t2
ON t1.column_name = t2.column_name; -- using aliases
```

Below is an example of an inner join on the `id` column. We assign simple aliases to the table names (a for `emp_list` and b for `score`).

```
%%sql
SELECT * FROM emp_list a JOIN score b ON a.id=b.id;
```

Running query in 'duckdb://'

	id	name	class	age	height	id_1	math	english	science
1	Yamada	1	14	150	1	10	10	40	
2	Tanaka	2	13	150	2	None	60	100	
3	Suzuki	1	13	150	3	30	40	60	
5	Ito	3	12	155	5	25	31	10	
6	Takeuchi	2	16	155	6	50	30	10	
8	Sato	1	14	160	8	90	100	90	

Now, let's do a left outer join using the `id` column. The table on the left is the base, and if no matching data exists on the right, it becomes None (NULL).

```
%%sql
SELECT * FROM emp_list a LEFT JOIN score b ON a.id=b.id;
```

Running query in 'duckdb://'

	id	name	class	age	height	id_1	math	english	science
1	Yamada	1	14	150	1	10	10	40	
2	Tanaka	2	13	150	2	None	60	100	
3	Suzuki	1	13	150	3	30	40	60	
5	Ito	3	12	155	5	25	31	10	
6	Takeuchi	2	16	155	6	50	30	10	
8	Sato	1	14	160	8	90	100	90	
4	Kato	2	15	150	None	None	None	None	None
7	Kimura	3	11	160	None	None	None	None	None

For reference, it is also possible to join a table to itself in what is called a self-join.

```
%%sql
SELECT * FROM emp_list a JOIN emp_list b ON a.id=b.id;
```

Running query in 'duckdb://'

id	name	class	age	height	id_1	name_1	class_1	age_1	height_1
1	Yamada	1	14	150	1	Yamada	1	14	150
2	Tanaka	2	13	150	2	Tanaka	2	13	150
3	Suzuki	1	13	150	3	Suzuki	1	13	150
4	Kato	2	15	150	4	Kato	2	15	150
5	Ito	3	12	155	5	Ito	3	12	155
6	Takeuchi	2	16	155	6	Takeuchi	2	16	155
7	Kimura	3	11	160	7	Kimura	3	11	160
8	Sato	1	14	160	8	Sato	1	14	160

[Key Points]

UNION: Concatenate multiple **SELECT** results

UNION ALL: Concatenate multiple **SELECT** results with duplicates

INNER JOIN: Inner join

LEFT OUTER JOIN: (Left) Outer join

✓ <Practice Question 7-12>

Create a new table called 'cardtb.' Include the columns id, point, and money (all INT).

Then insert the following data:

(id, point, money) = (1, 100, 1000), (3, 50, 500), (4, 30, 600), (5, 10, 10), (7, 100, 1000), (8, 2000, 100)

Start coding or [generate](#) with AI.

Start coding or [generate](#) with AI.

Start coding or [generate](#) with AI.

✓ <Practice Question 7-13>

Perform an inner join with the 'emp_list' table using id as the key.

Start coding or [generate](#) with AI.

✓ <Practice Question 7-14>

Perform an outer join with the 'emp_list' table using id as the key.

Start coding or [generate](#) with AI.

✓ 7.2.6 CASE Statements and Subqueries

✓ CASE Expression

When working with data, there are times when you might want to split it into categories and aggregate each one. SQL allows for conditional logic using **CASE** expression. The syntax is as follows:

```
CASE
  WHEN condition1 THEN result1
  WHEN condition2 THEN result2
  ...
  ELSE result
END;
```

In the following example, each class number is assigned a class name (className=A,B,C for class 1,2,3, respectively). The id, name, and class name are returned.

```
%%sql
SELECT id, name,
CASE
  WHEN class = 1 THEN 'A'
  WHEN class = 2 THEN 'B'
  WHEN class = 3 THEN 'C'
  ELSE 'others'
END AS className
FROM emp_list;
```

Running query in 'duckdb://'

id name className

1	Yamada	A
2	Tanaka	B
3	Suzuki	A
4	Kato	B
5	Ito	C
6	Takeuchi	B
7	Kimura	C
8	Sato	A

✓ Subquery

Subquery is a query inside a query, in a nested structure. Subqueries allow you to treat information extracted from the original table as a separate table.

In this example, we treat the contents of the `emp_list` table as a table named `b` using a subquery.

```
%%sql
```

```
SELECT b.name, b.age FROM
  (SELECT a.age, a.name FROM emp_list a WHERE a.class = 1) b;
```

Running query in 'duckdb://'

name age

Yamada	14
Suzuki	13
Sato	14

Note that in the above example, we specified column names, but if you want to select every column, you can write dot-asterisk (.) after the alias name.

```
%%sql
```

```
SELECT b.* FROM
  (SELECT a.* FROM emp_list a WHERE a.class = 1) b;
```

Running query in 'duckdb://'

id name class age height

1	Yamada	1	14	150
3	Suzuki	1	13	150
8	Sato	1	14	160

With this, we have finished covering the basic operations of SQL. After the practice questions, we will learn about views.

[Key Points]

CASE WHEN condition THEN A ELSE B END AS new_column_name:

Conditional branching

✓ <Practice Question 7-15>

In the 'emp_list' table, name any record whose height is below 155 as 'below_155,' exactly 155 as 'equal_155,' and above 155 as 'over_155.' Display the table.

Start coding or [generate](#) with AI.

✓ <Practice Question 7-16>

Using the above table and the subquery concept, find out how many people belong to each of those heightLevel categories (the column added in Practice Question 1).

Start coding or [generate](#) with AI.

✓ <Practice Question 7-17>

Using the above table and the subquery concept, find the average age for each of those heightLevel categories (the column added in Practice Question 1).

Start coding or [generate](#) with AI.

✓ 7.3 View

Keywords: View

✓ 7.3.1 View

Next, let's learn about views.

A **view** is a virtual table that stores your query logic in advance. So far, we have been querying data directly from tables to obtain the information we need. However, setting conditions and extracting data from each table every single time can be a bit tedious. We could create a table for every type of data we need, but this requires updating the table every time the source data changes. Using views allows us to always obtain the latest data easily without updating the table.

The method for creating a view is almost the same as for creating a table, except we use `CREATE VIEW`.

```
CREATE VIEW view_name AS
SELECT columns1, ...
FROM table_name;
```

Let's start with a simple example. Below, we create a view that retrieves only the `id` and `name` from the `emp_list` table.

```
%%sql
CREATE VIEW v1 AS
SELECT id, name
FROM emp_list;
```

Running query in 'duckdb://'
Count

Check whether the new view has been created. You can do this the same way you learned before.

```
%%sql
SELECT * FROM v1;
```

Running query in 'duckdb://'

id	name
1	Yamada
2	Tanaka
3	Suzuki
4	Kato
5	Ito
6	Takeuchi
7	Kimura
8	Sato

So far, this does not really demonstrate the convenience of views, so let's look at another example. By creating a view for a specific query, you would be able to access the results without having to write the query again.

```
%%sql
CREATE VIEW v2 AS
SELECT id, name,
CASE
    WHEN class = 1 THEN 'A'
    WHEN class = 2 THEN 'B'
    WHEN class = 3 THEN 'C'
    ELSE 'others'
END AS className
FROM emp_list;
```

Running query in 'duckdb://'